TEXORA · LECTURE SHEET

Chi-square Test

The χ^2 statistic comparing observed and expected frequencies, its properties and conditions, goodness of fit for a thrown die, and a 2 × 2 test of independence between training and performance — with degrees of freedom, critical values and Yates' correction.

STAT-2109 · PROBABILITY AND STATISTICS · TOPIC 11 · LEVEL 2 TERM 1

1. A test for frequencies, not for means

The z and t tests judge a mean or a proportion. Many practical questions, however, arrive as **counts**: how often each face of a die appears, how many workers fall into each combination of training and performance, how many fabric faults are of each type. For such data the instrument is the **chi-square test**, written χ^2-test after the Greek letter chi. A hypothesis H_0 tells us what frequencies to *expect* in each category if it is true; the data give the frequencies actually *observed*. Close agreement supports the hypothesis; disagreement beyond ordinary sampling variation rejects it. The χ^2 statistic measures the total disagreement as a single number whose sampling distribution is known and tabulated.

2. The χ^2 statistic

For categories i = 1, 2, …, k, let O_i be the observed frequency and E_i the expected frequency under H_0. Then

$$\chi^2 = \Sigma\,(O - E)^2 / E$$

the sum being taken over all categories. Note three features. Squaring prevents positive and negative discrepancies from cancelling. Dividing by E scales each discrepancy: a difference of 5 counts for far more against an expectation of 10 than against 500. And the expected frequencies are calculated from H_0 and the sample total, so $\Sigma E = \Sigma O = N$; being theoretical, they need not be whole numbers, as the second worked example shows.

The decision follows the pattern of Topics 9 and 10, with one difference of direction: χ^2 grows as the fit worsens. We therefore reject H_0 when the calculated χ^2 *exceeds* the table value at the chosen level and degrees of freedom. Small values of χ^2, close to zero, mean close agreement between observation and hypothesis.

3. Properties of the χ^2 distribution

- **Always non-negative.** Since χ^2 is a sum of squared quantities divided by positive expected frequencies, $\chi^2 \geq 0$ always. The value zero occurs only when every observed frequency exactly equals its expectation.
- **Right-skewed shape.** The distribution is not symmetric. It starts at zero, rises to a peak, and has a long right tail. The test is therefore inherently one-sided in the statistic: only large values count against H_0.
- **Depends on degrees of freedom.** As with t, there is a family of χ^2 curves indexed by df. As df increases the peak moves to the right and the curve flattens and becomes more nearly symmetric.
- **Additive.** Independent χ^2 variables add: the sum of independent χ^2 statistics is itself χ^2, with degrees of freedom equal to the sum of the separate degrees of freedom. This property lets results from independent tables or experiments be combined.
- **Mean equal to df.** Under H_0 the expected value of the statistic is its degrees of freedom, a useful rough check: a calculated χ^2 near its df is unremarkable, while one several times larger invites rejection.

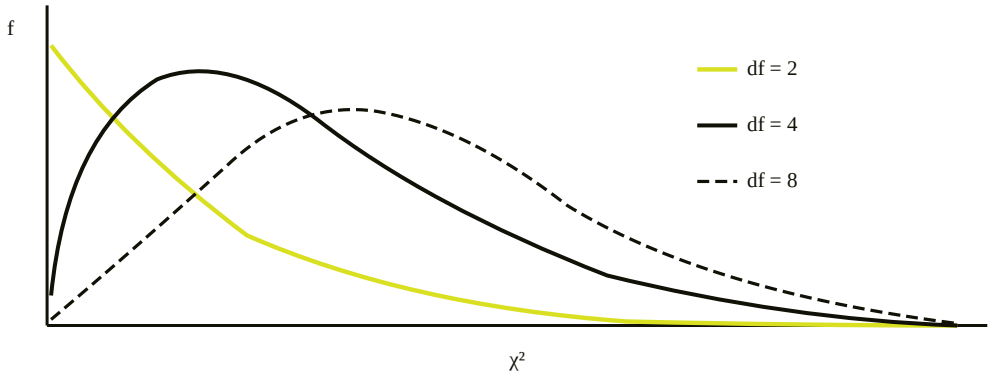


Figure 1 — Chi-square distributions for df = 2, 4 and 8: right-skewed curves whose peak moves right and flattens as df increases

4. Conditions for the χ^2 test

The tabulated probabilities of χ^2 are approximations that hold only when the data meet the following conditions. They should be checked before any calculation is trusted.

- **Total frequency large enough:** the grand total N should be at least 50.
- **Expected frequencies not too small:** no expected frequency E should be less than 5. Where a category has a small expectation, it should be combined with a neighbouring category (pooling) before the test is applied, with the degrees of freedom reduced accordingly.

- **Independent observations:** each item counted must be independent of the others; one observation must not influence the category into which another falls.
- **Raw counts, not percentages:** the statistic must be computed from actual frequencies. Percentages, proportions or ratios destroy the information about sample size on which the distribution depends, and give wrong results.
- **Unrestricted categories drawn by random sampling:** the observations should come from a random sample, classified into the stated categories without further restriction beyond the fixed total N (and any parameters estimated from the data, which reduce df).

5. Degrees of freedom and critical values

The degrees of freedom count the categories that are free to vary once the totals and any estimated parameters are fixed.

- **Goodness of fit:** df = n − 1, where n is the number of categories (often written k − 1). If parameters are estimated from the data, one further degree of freedom is lost for each.
- **Contingency table with r rows and c columns:** df = (r − 1)(c − 1). For a 2 × 2 table this gives df = 1.

DEGREES OF FREEDOM	CRITICAL VALUE AT 5%	CRITICAL VALUE AT 1%
1	3.841	6.635
4	9.488	—
5	11.070	15.086

The two values used in this sheet, 3.841 for df = 1 and 11.070 for df = 5, both at the 5% level, are worth remembering. As always, a calculated χ^2 above the table value leads to rejection of H_0; below it, H_0 is not rejected.

6. Worked example 1: goodness of fit — is a die fair?

A die is thrown 60 times, with the results below. Under H_0, that the die is fair, each face has probability 1/6, so the expected frequency of each face is 60 / 6 = 10.

FACE	1	2	3	4	5	6	TOTAL
Observed O	8	12	9	11	10	10	60
Expected E	10	10	10	10	10	10	60

$(O - E)^2 / E$	0.4	0.4	0.1	0.1	0	0	$\chi^2 = 1.0$

WORKED EXAMPLE

Hypotheses. H_0: the die is fair, each face having probability 1/6; H_1: the die is not fair. Level 5%.

Terms. $(8 - 10)^2 / 10 = 0.4$; $(12 - 10)^2 / 10 = 0.4$; $(9 - 10)^2 / 10 = 0.1$; $(11 - 10)^2 / 10 = 0.1$; and 0 for each of the last two faces.

Statistic. $\chi^2 = 0.4 + 0.4 + 0.1 + 0.1 + 0 + 0 = 1.0$.

Degrees of freedom and critical value. df = 6 − 1 = 5; the table value at the 5% level is 11.070.

Decision. 1.0 < 11.070, so the disagreement between observed and expected frequencies is well within chance variation. Accept H_0: the die is fair, as far as this evidence goes.

Goodness of fit is the general name for this use: a theoretical distribution, here the uniform distribution over six faces, is fitted to the data and the closeness of fit judged. The same procedure tests whether faults follow a Poisson pattern, whether grouped measurements follow a normal distribution, or whether quality counts follow a stated ratio.

7. Worked example 2: test of independence in a 2 × 2 table

One hundred workers are classified by whether they received training and whether their performance is satisfactory. H_0 states that training and satisfactory performance are **independent**, that is, unassociated.

	SATISFACTORY	NOT SATISFACTORY	ROW TOTAL
Trained	30	20	50
Untrained	15	35	50
Column total	45	55	N = 100

Under independence, the expected frequency of a cell is (row total × column total) / N. Hence the expected frequencies are (50 × 45) / 100 = 22.5, (50 × 55) / 100 = 27.5, (50 × 45) / 100 = 22.5 and (50 × 55) / 100 = 27.5, in the same positions as 30, 20, 15 and 35. Every $|O - E|$ in this table is 7.5.

WORKED EXAMPLE

$\chi^2 = (7.5)^2 / 22.5 + (7.5)^2 / 27.5 + (7.5)^2 / 22.5 + (7.5)^2 / 27.5$

$= 2.5 + 2.045 + 2.5 + 2.045 =$ **9.09**.

Check by the 2 × 2 shortcut. With cells a = 30, b = 20, c = 15, d = 35: $\chi^2 = N(ad - bc)^2 / [(a + b)(c + d)(a + c)(b + d)] = 100 \times (30 \times 35 - 20 \times 15)^2 / (50 \times 50 \times 45 \times 55) = 100 \times 750^2 / 6{,}187{,}500$

= 9.09 ✓, agreeing with the direct sum.

Degrees of freedom and decision. df = (2 − 1)(2 − 1) = 1; the critical value at the 5% level is 3.841. Since 9.09 > 3.841, reject the hypothesis of independence: training and satisfactory performance are **associated**. The observed pattern, trained workers satisfactory more often than expected and untrained workers less often, is too strong to attribute to chance at the 5% level.

Two cautions belong with this conclusion. Association is not causation: trained workers might also differ in experience or selection. And the result is in fact stronger than the 5% statement: even at the 1% level, whose critical value is 6.635, 9.09 still leads to rejection.

8. Yates' correction for continuity

In a 2 × 2 table df is only 1 and the counts are discrete, while the χ^2 distribution is continuous, so the approximation can overstate the statistic slightly. **Yates' correction** reduces each |O − E| by 0.5 before squaring, using $(|O - E| - 0.5)^2 / E$. It makes the test slightly more conservative and is applied only to 2 × 2 tables, never to larger tables or many-category fits. In the worked example above, 9.09 is so far above 3.841 that the correction would not alter the decision.

9. Applications and limitations

Applications. The three standard uses are: **goodness of fit**, judging whether observed frequencies follow a stated theoretical distribution; **independence**, judging whether two classifications in a contingency table are associated; and **homogeneity**, judging whether several populations share the same distribution over a set of categories, computed by exactly the same arithmetic as the independence test. In textile work these cover fault-type counts against an expected pattern, operator or shift classifications against quality grades, and comparisons of defect profiles across mills or machines.

Limitations. The test works only on frequencies, and its verdict is sensitive to sample size: with a very large N, trivial departures from H_0 become significant, while with a small N real departures may be missed. It says that disagreement exists, but not where it lies or how large it is practically; the individual (O − E) values must be inspected for that. Pooling to rescue small expected frequencies changes the degrees of freedom and must be decided before seeing the result. Above all, a significant χ^2 establishes association, not cause.

Glossary

TERM	MEANING

Observed frequency (O)	The count actually found in a category in the sample data.
Expected frequency (E)	The count predicted for a category by H_0, calculated so that ΣE equals the sample total N.
χ^2 statistic	$\Sigma (O - E)^2 / E$, summed over all categories; it measures the total disagreement between data and hypothesis.
Goodness of fit	The use of χ^2 to judge how closely observed frequencies follow a theoretical distribution.
Contingency table	A two-way table of frequencies classified by two attributes, with r rows and c columns.
Independence	The null hypothesis that two classifications are unassociated, giving E = (row total × column total) / N.
Degrees of freedom	For fit, categories minus 1; for a contingency table, $(r - 1)(c - 1)$; 1 for a 2 × 2 table.
Yates' correction	Reducing each $\|O - E\|$ by 0.5 before squaring, applied only in 2 × 2 tables to correct for continuity.
Pooling	Combining categories with expected frequencies below 5 so that the χ^2 approximation remains valid.

Key points & summary

- $\chi^2 = \Sigma (O - E)^2 / E$ compares observed frequencies with those expected under H_0; large values count against H_0.
- χ^2 is always ≥ 0, right-skewed, additive, and governed by its degrees of freedom.
- Conditions: $N \geq 50$, every $E \geq 5$ (pool if necessary), independent observations, and raw counts rather than percentages.
- Degrees of freedom: categories − 1 for goodness of fit; $(r - 1)(c - 1)$ for a contingency table.
- Critical values: df = 1 gives 3.841 at 5% and 6.635 at 1%; df = 5 gives 11.070 at 5% and 15.086 at 1%.
- Die example: $\chi^2 = 1.0$ with df = 5, well below 11.070, so the die is judged fair.
- Training example: $\chi^2 = 9.09$ with df = 1, above 3.841, so training and satisfactory performance are associated; association alone does not prove causation.

Practice

Answer in your own words; brief model answers follow.

1. A goodness-of-fit test has 5 categories and no estimated parameters. State the degrees of freedom and the 5% critical value, using the table in this sheet where possible, and say what extra table information a test with 4 degrees of freedom requires.
2. Explain why each term of χ^2 is divided by E, and why the test must be computed from counts rather than percentages.
3. A die is thrown 60 times and a face expected 10 times is observed 16 times. Compute that face's contribution to χ^2, and explain why one large contribution alone does not settle the test.
4. For a 3×4 contingency table, compute the degrees of freedom. If the calculated χ^2 is large, state the conclusion in words.
5. In the training and performance example, verify the expected frequency for "Untrained and satisfactory", and explain why the expected frequencies need not be whole numbers.
6. State Yates' correction, when it is used, and one limitation of the χ^2 test concerning cause and effect.

Answer key

1. df = 5 − 1 = 4. The 5% critical value for df = 4 is 9.488, given in the table of Section 5. No extra information is needed beyond that table value and the calculated statistic; the decision is to reject H_0 if the calculated χ^2 exceeds 9.488.
2. Dividing by E scales each discrepancy by what was expected: a difference of a given size is more surprising against a small expectation than a large one. Percentages discard N, and since the sampling distribution of χ^2 depends on the actual counts, percentages would give a statistic with no valid table probabilities.
3. Contribution = $(16 - 10)^2 / 10 = 36 / 10 = 3.6$. The test judges the sum over all faces against the table value for df = 5, 11.070; the remaining faces also contribute, and only the total decides.
4. df = $(3 - 1)(4 - 1) = 2 \times 3 = 6$. A χ^2 above the table value for 6 df means the two classifications are not independent; they are associated in the population sampled.
5. E = (row total × column total) / N = $(50 \times 45) / 100 = 22.5$. Expected frequencies are theoretical averages under H_0 over many repetitions, not actual counts in any single sample, so fractions are natural and correct.
6. Yates' correction replaces $|O - E|$ by $|O - E| - 0.5$ before squaring; it is used only for 2×2 tables (df = 1), where the discrete counts meet a continuous distribution. Limitation: a significant χ^2 shows association between classifications, not that one causes the other.